Illustrative only: a real 2014 background recording with a simulated impact sound at the predicted level

real background + simulated impact sound
simulated impact sound alone
time at which the simulated pulse is inserted

coupling efficiency 0.001
median predicted level

coupling efficiency 0.001
90th percentile predicted level

coupling efficiency 0.00021 (F-35A, provisional)
median predicted level

coupling efficiency 0.00021 (F-35A, provisional)
90th percentile predicted level

Perth Canyon logger 3376 (real 2014 background)
Pressure (Pa)

level 91 dB re 1 µPa, -11 dB above the noise median
matched-filter peak 0.36 (threshold 0.897: below)

level 103 dB re 1 µPa, +1 dB above the noise median
matched-filter peak 0.68 (threshold 0.897: below)

level 84 dB re 1 µPa, -18 dB above the noise median
matched-filter peak 0.36 (threshold 0.897: below)

level 96 dB re 1 µPa, -5 dB above the noise median
matched-filter peak 0.36 (threshold 0.897: below)

Cape Leeuwin stand-in (3376 background +5.4 dB)
Pressure (Pa)

level 100 dB re 1 µPa, -7 dB above the noise median
matched-filter peak 0.37 (threshold 0.897: below)

level 115 dB re 1 µPa, +8 dB above the noise median
matched-filter peak 0.95 (threshold 0.897: above)

level 93 dB re 1 µPa, -14 dB above the noise median
matched-filter peak 0.37 (threshold 0.897: below)

level 108 dB re 1 µPa, +1 dB above the noise median
matched-filter peak 0.52 (threshold 0.897: below)

Diego Garcia South stand-in (3376 background -0.7 dB)
Pressure (Pa)

level 91 dB re 1 µPa, -10 dB above the noise median
matched-filter peak 0.36 (threshold 0.897: below)

level 105 dB re 1 µPa, +4 dB above the noise median
matched-filter peak 0.86 (threshold 0.897: below)

level 84 dB re 1 µPa, -17 dB above the noise median
matched-filter peak 0.36 (threshold 0.897: below)

level 98 dB re 1 µPa, -3 dB above the noise median
matched-filter peak 0.35 (threshold 0.897: below)

Time in the recording (s)

Plain note: Each panel shows 140 seconds of a real sound recording from the Perth Canyon logger on 8 March 2014 (blue), with a simulated impact sound added at 60 seconds. The red trace is the simulated impact sound alone. Its size is the median or the 90th-percentile sound level that the simulated impacts would give at that station, for two assumed shares of impact energy that become sound. The middle and lower rows use the same recording, made louder or quieter to match the noise at Cape Leeuwin and Diego Garcia South. This is an illustration. It is not a recording of an aircraft impact and it is not a detection.

Technical note: Background: IMOS 3376 recording MH370/3376_PerthCanyon/531A5E9E.DAT (logger clock 2014-03-08 00:05:02, role B; role-S recordings never opened), Curtin gain file + sensitivity calibration, 5–40 Hz zero-phase 4th-order Butterworth, 140 s from t = 60 s; its 1 s mean-square median is 101.5 dB re 1 uPa (the detectable shares in charts 1–3 use the 14-day background median, 96.2 dB; this recording is louder). Pulse: the weak-signal track's KRAKEN dispersed template (templates_raw.npz h_2376.1 / h_2200.0 / h_3703.5 km, resampled to 500 Hz, same filter) scaled so its peak 1 s mean square equals the predicted 5–40 Hz level (+4.8 dB triad gain at H01W/H08S, as an array-equivalent level). Levels: run J D0′ (00:19 held out), p = 0.5, weighted median and 90th percentile of the 5–40 Hz 1 s-equivalent total, recomputed tonight with the corrected IMOS transmission-loss table (imos_tl_ram_arc.csv; reproduces levels-rerun-v2 to 0.01 dB); IMS loss unchanged. Stand-ins: 3376 background + Blackman App. B offset (H01W +5.4 dB, H08S −0.7 dB, digitised). Matched-filter statistic = normalised cross-correlation of the 0.5 s squared envelope with the template envelope (M1b-type); false-alarm threshold 0.897 (alpha 0.01, 3376 null, weak-signal note). Timing on the x axis is time in the recording, not UTC. PREDICTIVE, NOT CALIBRATED. C1: propagation slope misfit +10.4 / +7.4 dB per octave at H01W / H08S (after the EW0112 source result a source-term caveat; air9 verdict NOT EXPLAINED, gap G23). C2: coupling efficiency eta not calibrated; swept over four decades; F-35A upper anchor eta = 2.1e-4 is provisional (from Brown 2026's quote of Metz 2023; re-derivation from Metz pending). No detection is claimed.